

Sam Lee

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EDUCATION

University of Wisconsin - Madison

Bachelor of Science — Computer Science

Madison, WI

Sep. 2023 – Dec. 2025

- **CS Coursework:** Object Oriented Programming, Machine Organization, Intro to AI, Big Data Systems, Database Systems, UI/UX, Algorithms and Data Structures

Madison Area Technical College

Transfer Coursework

Madison, WI

Aug. 2018 – May 2020

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Java, Python, SQL, HTML/CSS

Frameworks/Libraries: React, Next.js, Express, Tailwind CSS, shadcn/ui

Cloud/Deployment: AWS (EC2, S3, Route 53) Nginx, PM2

Databases: MySQL

Developer Tools: Git, Postman, Jira, Slack

PROJECTS

Swing Analyzer | *React, Express, shadcn, MySQL, Nginx, AWS, Gemini API* | [Live Demo](#) Aug. 2026 – Present

- Developing a full-stack application that will allow golfers to upload swing videos and receive personalized AI-generated feedback.
- Implemented user authentication and protected API endpoints using JWT authentication, HTTP-only cookies, password hashing, and MySQL.
- Built a layered Express backend and deployed the application on AWS EC2 using Nginx, HTTPS, and PM2 for secure routing and automatic process recovery.
- Implemented direct-to-S3 video uploads using presigned URLs and integrated Gemini API to generate personalized golf swing feedback.

Personal Portfolio Website | *NextJS, Tailwind CSS, shadcn, OpenAI API* | [Live Demo](#) Jul. 2026 – Jul. 2026

- Developed a responsive portfolio website using Next.js and TypeScript featuring an AI-powered chatbot, project showcase, and blog section.
- Integrated the OpenAI API to build a chatbot that retrieves portfolio information and answers questions about projects, technical skills, and professional experience.
- Deployed the application on Vercel for production hosting and continuous deployment.

UW-Madison Capstone Team Project | *TypeScript, React, Tailwind CSS* | [Live Demo](#) Sep. 2025 – Dec. 2025

- Developed reusable React components and responsive user interfaces for product discovery, ranking, and leaderboard pages using TypeScript, React, and Tailwind CSS.
- Built modular frontend components and responsive layouts, improving code maintainability and providing a consistent user experience across multiple application pages.
- Collaborated in an Agile development workflow using GitHub, Jira, and Slack, contributing to feature development, debugging, and version-controlled team collaboration.

WORK EXPERIENCE

Machine Vision Inspection Technician

LSP USA LLC — Ultium Cells Battery Plant

May 2026 – Present

Spring Hill, TN

- Providing technical support for machine vision inspection systems used in automated battery manufacturing.
- Monitoring machine vision inspection systems across production lines, review flagged images to distinguish actual defects from false positives, adjust inspection parameters to reduce over-detection, and promptly notify production teams of confirmed defects.
- Diagnosing hardware and software issues, identify root causes, and implement solutions to restore equipment performance and minimize production downtime.
- Collaborating with production, maintenance, and engineering teams by communicating equipment issues and troubleshooting updates in real time, enabling rapid corrective action and improved system reliability.